

DEPTH-FOUR HOLE REFILL IS DRY ON A 2969-WORD (23, 6, 10) CONSTANT-WEIGHT CODE

ABSTRACT. Let C_0 be the 2969-word binary constant-weight code **a23.6.10.2969H** of length 23, weight 10 and minimum distance 6, listed in full in Appendix A. Lysenstoen [3] proves that no exchange deleting at most three words of C_0 and inserting more words than it deletes produces a valid (23, 6, 10) code, and states the depth-4 analogue as an open problem. We report an exhaustive verification that no such exchange exists at depth 4 either. An elementary reduction, the one described in [2, §XII], cuts the $\binom{2969}{4} = 3\,231\,108\,527\,626$ four-element deletion sets to 4 948 069, which with the smaller sizes leaves a space of 5 094 818 sets; the 14 447 of those that pass a necessary counting test are each decided by an exact maximum-clique computation, and the remaining 5 080 371 by that count alone. For every deletion set D in that space the largest pairwise-compatible set of insertable words has size at most $|D|$, and that bound is attained for each of $|D| = 1, 2, 3, 4$. Since [2, §XII] reports achieving k -optimality for k in the range 2–5 without recording which codes it certified, we claim no priority at any depth; and since C_0 has been superseded, no lower bound on $A(23, 6, 10)$ changes.

1. THE PROBLEM

Points are $1, \dots, 23$ and a *word* is a 10-element subset of them, written as a 23-bit mask. Two words u, v satisfy $d(u, v) = 2(10 - |u \cap v|)$, so

$$(1) \quad d(u, v) \geq 6 \iff |u \cap v| \leq 7,$$

and we call such a pair *compatible*. A (23, 6, 10) *code* is a set of pairwise distinct, pairwise compatible words; $A(23, 6, 10)$ is the largest size of one.

Fix

$$C_0 = \mathbf{a23.6.10.2969H},$$

the 2969-word (23, 6, 10) code from Brouwer’s table [1], reproduced in full in Appendix A. The SHA-256 digest of the in-order concatenation of its 2969 six-hex-digit words is

$$2\text{cfc5d1bd8e2e61dc73a2ab31e3ec985d268d67cc635468912828c8a21b5cbfd}.$$

A *candidate* is a word that is not in C_0 ; there are $\binom{23}{10} - 2969 = 1\,141\,097$ of them. The *blockers* of a candidate s are

$$B(s) = \{c \in C_0 : |s \cap c| \geq 8\},$$

2020 *Mathematics Subject Classification*. 94B25, 94B65, 05B40.

Key words and phrases. binary constant-weight code, $A(23, 6, 10)$, local exchange, hole refill, exhaustive search, maximum clique.

the words of C_0 that s is *not* compatible with, and for $D \subseteq C_0$ we put

$$\text{Cand}(D) = \{s \text{ a candidate} : B(s) \subseteq D\}.$$

So $\text{Cand}(D)$ is exactly the set of words that may legally be inserted once D has been deleted. All of this is the vocabulary of [3].

The statement under test. Lysenstoen proves [3, the depth-3 theorem, `main.tex` lines 196–200 of `arXiv:2607.19550v1`]:

“There is no pair (D, S) with $D \subseteq C_0$, $|D| \leq 3$, and S a set of weight-10 words, such that $(C_0 \setminus D) \cup S$ is a valid $(23, 6, 10)$ code with $|S| > |D|$.”

and then states, in a paragraph of its own carrying no number (`main.tex` lines 221–225 of the same e-print):

“Open problem. The depth-4 analogue is unresolved: the model (30,247 candidates, $\sim 888\text{k}$ pre-seeded pairs) was not closed within the budgets tried (CP-SAT, CBC, and SCIP each left it open), so we state it as an open problem rather than a theorem.”

The target carries no statement number, so it is quoted in full; the line numbers are those of the L^AT_EX e-print.

Remark (the implicit disjointness clause). Read literally both statements are false, the hypothesis $S \cap C_0 = \emptyset$ having been left implicit: take $|D| = 1$ and let S be any two words of $C_0 \setminus D$. Insertion of candidates only is the intended reading, since [3] defines a candidate to be a weight-10 word not in C_0 and the insertion variables of its model range over candidates only. We therefore carry the clause as a hypothesis in Theorem 1.

Theorem 1. *Let C_0 be the code of Appendix A. There is no pair (D, S) with $D \subseteq C_0$, $|D| \leq 4$, and S a set of pairwise distinct words with $S \cap C_0 = \emptyset$, such that $(C_0 \setminus D) \cup S$ is a $(23, 6, 10)$ code and $|S| > |D|$.*

Theorem 2. *Let $\mathcal{D}$ be the family of all $D \subseteq C_0$ with $|D| \leq 4$ that are unions of at least one and at most four blocker sets; $|\mathcal{D}| = 5\,094\,818$. For every $D \in \mathcal{D}$ the largest pairwise-compatible subset of $\text{Cand}(D)$ has size at most $|D|$; over the $4\,948\,069$ members with $|D| = 4$ one has $|\text{Cand}(D)| \leq 6$. The bound $|D|$ is attained at each of $|D| = 1, 2, 3, 4$.*

Theorem 2 implies Theorem 1 through Lemmas 3–4.

2. THE REDUCTION

Lemma 3. *Let (D, S) be as forbidden in Theorem 1, and put $D' = \bigcup_{s \in S} B(s)$. Then $D' \subseteq D$, the pair (D', S) is again of that form, and D' is a union of at most $|D'| \leq 4$ blocker sets. Hence $D' \in \mathcal{D}$.*

Proof. Each $s \in S$ is compatible with every word of $C_0 \setminus D$ that survives beside it, so $B(s) \subseteq D$ and therefore $D' \subseteq D$. Every $s \in S$ still has

$B(s) \subseteq D'$, so no word of $C_0 \setminus D'$ conflicts with any inserted word, and the words of S are pairwise compatible because they already were; thus $(C_0 \setminus D') \cup S$ is a valid code. Finally $|S| > |D| \geq |D'|$, so the pair is again of the forbidden form. For the last clause, take a subfamily of $\{B(s) : s \in S\}$ of minimum size whose union is D' and discard any empty member; each remaining member contributes at least one point of D' not contributed by the others, so there are at most $|D'| \leq |D| \leq 4$ of them. There is also at least one: $S \neq \emptyset$ because $|S| > |D| \geq 0$, and no blocker set is empty, since a candidate with no blocker could be adjoined to C_0 , whereas C_0 is maximal — a computed fact, recorded in §3. So $D' \neq \emptyset$, as every member of $\mathcal{D}$ is. $\square$

Lemma 4. *A pair as forbidden in Theorem 1 exists if and only if some $D \in \mathcal{D}$ admits $|D| + 1$ pairwise compatible members of $\text{Cand}(D)$.*

Proof. ($\Rightarrow$) Let (D, S) be such a pair and D' as in Lemma 3. Then $S \subseteq \text{Cand}(D')$, S is pairwise compatible, and $|S| \geq |D'| + 1$; take any $|D'| + 1$ members. ($\Leftarrow$) Let $S' \subseteq \text{Cand}(D)$ be pairwise compatible with $|S'| = |D| + 1$ and put $D'' = \bigcup_{s \in S'} B(s) \subseteq D$. Every word of $C_0 \setminus D''$ is compatible with every member of S' , the members of S' are pairwise compatible and lie outside C_0 , so $(C_0 \setminus D'') \cup S'$ is a $(23, 6, 10)$ code of size $2969 - |D''| + |S'| \geq 2970$, and $|S'| = |D| + 1 > |D| \geq |D''|$. $\square$

So the question is a finite one about $\mathcal{D}$, and $\mathcal{D}$ is small: the reduction replaces $\binom{2969}{4} = 3\,231\,108\,527\,626$ four-element deletion sets by the 4 948 069 of them that lie in $\mathcal{D}$, a factor 6.53×10^5 . Lemma 3 uses no theorem of [3], only the definitions, so pruning with it cannot import an error from the statement being tested. Neither the reduction nor the vocabulary is new. Already [2, §XII] forms the bipartite graph with “a blue vertex for every codeword of C and a red vertex for every vector not in C that ‘bites’ (lies at distance less than d from) k or fewer words of C ”, and then, for each k -subset S of C , searches “among the red vertices that are connected only to S ” — in the notation of §1, the candidates s with $B(s) \subseteq D$.

Two further facts make $\mathcal{D}$ computable. First, every blocker set occurring in a member of $\mathcal{D}$ has at most four elements, so the members of $\mathcal{D}$ are exactly the unions of at least one and at most four elements of the finite seed family

$$\mathcal{B} = \{B(s) : s \text{ a candidate, } |B(s)| \leq 4\}, \quad |\mathcal{B}| = 30\,117,$$

carried by the 30 247 candidates with at most four blockers. Second, since every intermediate union in such a chain is a subset of the final one, $\mathcal{D}$ is the closure of $\mathcal{B}$ under the partial operation “union, when the result still has at most four elements”. Computing that closure is a fixed-point computation on a family of about five million four-element subsets of a 2969-element set.

3. THE COMPUTATION

The blocker census. The census of $B(s)$ over all 1 141 097 candidates is the only step that touches the whole candidate space, and it was done

twice by disjoint methods. Method 1 enumerates, for each $c \in C_0$, the $130 + 3510 = 3640$ words meeting c in 9 or in 8 points and accumulates the resulting multiset; this is sound because a word meeting c in 10 points equals c . Method 2 uses that, for words of weight 10, $|s \cap c| \geq 8$ holds if and only if c contains one of the $\binom{10}{8} = 45$ eight-element subsets of s : it indexes the $2969 \cdot 45 = 133\,605$ eight-element subsets of codewords and then sweeps all $\binom{23}{10} = 1\,144\,066$ masks. The two methods agree on all 30 247 pairs $(s, B(s))$ with $|B(s)| \leq 4$, and Method 2, which visits every mask explicitly, certifies in addition that

$$\#\{s : B(s) = \emptyset\} = 0,$$

i.e. C_0 is a maximal $(23, 6, 10)$ code. The distribution of $|B(s)|$ over the small values is

k	1	2	3	4
$\#\{s : B(s) = k\}$	70	1178	6503	22496
$\#\{s : B(s) \leq k\}$	70	1248	7751	30247

Calibration against the source. The counts 1 141 097, 1248, 7751, 30 247 and the absence of zero-blocker candidates are published in [3] and are reproduced above; so are its three pair counts, which pin our hex-to-bit reading of the listing: 5261 conflicting pairs among the 1248 candidates with at most two blockers, and 43 105 and 887 964 pairs that both conflict and share a blocker among the 7751 and among the 30 247, the last being the “~888k” of the open problem.

The deletion space. The closure described in §2 terminates after three rounds and

$$|\mathcal{D}| = \underbrace{70}_{|D|=1} + \underbrace{3567}_{|D|=2} + \underbrace{143\,112}_{|D|=3} + \underbrace{4\,948\,069}_{|D|=4} = 5\,094\,818.$$

That the result is the least fixed point was verified directly: every member is the union of the blocker sets it contains, and adjoining any one member of $\mathcal{B}$ to any of the 146 749 members of size at most three produces nothing outside the family.

The sweep. For each of the 5 094 818 members, $\text{Cand}(D)$ was assembled as $\bigcup_{T \subseteq D} \{s : B(s) = T\}$. When $|\text{Cand}(D)| \leq |D|$ no subset of $\text{Cand}(D)$ can have more than $|D|$ elements and nothing further is needed; otherwise the exact maximum clique of the compatibility graph on $\text{Cand}(D)$ was computed by branch and bound under a cardinality bound. No shard, no time limit and no heuristic cutoff were used; the largest instance has six vertices. Over the 4 948 069 members with $|D| = 4$,

j	1	2	3	4	5	6
$\#\{D : \text{Cand}(D) = j\}$	17 027	936 347	2 791 429	1 189 193	13 911	162

which sums to 4 948 069; in particular $|\text{Cand}(D)| \leq 6$ always, and only $13\,911 + 162 = 14\,073$ of the four-element deletion sets carry as many as five

candidates. The condition $|\text{Cand}(D)| > |D|$ is necessary for a witness but not sufficient — what Lemma 4 asks for is a $(|D| + 1)$ -clique — and exactly 14 447 members satisfy it. Each of those received a full exact maximum-clique search, and that search, not the count, is what decided it. The joint distribution of $(|D|, \omega(\text{Cand}(D)))$ over those 14 447 members is

ω	1	2	3	4
$ D = 2$	2	3		
$ D = 3$	7	154	208	
$ D = 4$	16	753	6179	7125

which sums to 14 447, and whose $|D| = 4$ row sums to $16+753+6179+7125 = 14\,073$ — exactly the count of four-element members with $|\text{Cand}(D)| \geq 5$ obtained from the other table, so two independently accumulated tallies agree. No entry lies to the right of the diagonal $\omega = |D|$. By Lemma 4 this proves Theorem 1, and the two tables together prove Theorem 2.

4. ATTAINMENT

Each deletion set counted on the diagonal $\omega = |D|$ of the last table admits a break-even exchange of $|D|$ words for $|D|$ words; those entries are lower bounds on the number of such sets, since only the 14 447 members with $|\text{Cand}(D)| > |D|$ were clique-searched. We exhibit one member of $\mathcal{D}$. Let D consist of the words at 0-based positions 27, 31, 585, 1333 of Appendix A, namely

$$D = \{294\text{CE8}, 29630\text{D}, 463\text{A0D}, 623607\}.$$

Then $\text{Cand}(D)$ has exactly five members, whose blocker sets are as shown:

s	$B(s)$ (positions)
0B630D	{31}
214D69	{27}
423E0D	{585, 1333}
4A3705	{1333}
4E3305	{585}

so $|\text{Cand}(D)| = 5 > 4 = |D|$ and D is one of the 14 447 live members. The ten pairwise intersections inside $\text{Cand}(D)$ are

$ \cdot \cap \cdot $	0B630D	214D69	423E0D	4A3705	4E3305
0B630D	—	5	6	7	7
214D69		—	4	3	2
423E0D			—	8	7
4A3705				—	9
4E3305					—

Every entry except the two in bold is at most 7. By (1) the only illegal pairs are $\{423\text{E0D}, 4\text{A3705}\}$ and $\{4\text{A3705}, 4\text{E3305}\}$; both contain **4A3705**, so

removing that one word leaves a 4-clique and no 5-clique can exist. Hence the maximum clique is

$$S_4 = \{0B630D, 214D69, 423E0D, 4E3305\},$$

of size exactly $4 = |D|$, and it is the unique one.

Deleting the positions listed and inserting the words listed now gives valid $(23, 6, 10)$ codes of size 2969 at four depths:

k	delete (positions)	insert
4	27, 31, 585, 1333	0B630D, 214D69, 423E0D, 4E3305
3	27, 585, 1333	214D69, 423E0D, 4E3305
2	585, 1333	423E0D, 4E3305
1	27	214D69

In each case the gain $|S| - |D|$ is exactly 0 and the resulting code again has minimum distance exactly 6. The case $k = 1$ is a direct consequence of the census: all 70 one-element members of $\mathcal{D}$ have $|\text{Cand}(D)| = 1$, so each yields a 1-for-1 break-even exchange and none yields a 2-for-1.

5. SCOPE

Neither the move nor the reduction that makes it searchable is ours. [2, §XII] calls a code *k-optimal* “if its cardinality cannot be increased (while maintaining the minimal distance) by deleting k words and adding holes”, gives the reduction quoted in §2, and reports that “in the range of our tables we are able to achieve k -optimality for values of k ranging 2–5”, a range containing 4. It records no list of which codes were so certified, and its Table XVI, the listing of codes with no known structure, passes from $(23, 6, 8)$ to $(28, 6, 5)$ with no $(23, 6, 10)$ entry. We have therefore not determined whether that 1990 certification covers the particular 2969-word listing C_0 of Appendix A, and we claim no priority for any layer of Theorem 1: what this note supplies for C_0 is a complete and independently checkable search, not a new method, and a reader should treat the $|D| = 4$ layer as possibly covered in [2, §XII]. Independently of that, the $|D| \leq 3$ layer of $\mathcal{D}$ — $70 + 3567 + 143\,112 = 146\,749$ members — re-proves the depth-3 theorem of [3] quoted above, and within the $|D| = 4$ layer [3] proves a depth-unbounded statement excluding every S all of whose members carry at most two blockers.

C_0 has been superseded: [3] proves $A(23, 6, 10) \geq 2979$, and Brouwer’s table [1] recorded $A(23, 6, 10) \geq 2992$ when we read it on 30 August 2026. Nothing here improves or weakens any entry of that table. Not examined: depth 5 and above, and codes other than C_0 . One limitation is internal to the accompanying program: in its histogram of $\omega(\text{Cand}(D)) - |D|$ over all 5 094 818 members, the buckets $-2, -1, 0$ are upper-bound-derived rather than measured, because for the 5 080 371 members with $|\text{Cand}(D)| \leq |D|$ the quantity recorded is $\min(|\text{Cand}(D)|, |D|) - |D|$ and no clique search was run. Nothing in Theorem 1 or 2 depends on that distinction, a clique being unable to exceed $|\text{Cand}(D)|$.

6. REPRODUCTION

The computation is determined by the 2969 words of Appendix A and the definitions of §1; there is no random choice, no external solver and no external data anywhere in it. It is a single pass over the $\binom{23}{10}$ masks followed by five million assemblies of $\text{Cand}(D)$, of which the 14447 that pass the counting test call for a clique computation on a graph with at most six vertices. The attainment claims of §4 need no machine: the five words of $\text{Cand}(D)$, their blocker sets and the ten intersections are printed above.

The program accompanying this note does exactly that, in Python with no third-party package, reading the listing of Appendix A as its only input and re-deriving every integer printed in §§1–4 above, including the eight taken from [3]. It runs in about a minute and a half on one core and reports 39 checks, all passing, followed by its own list of what it does not cover.

APPENDIX A. THE CODE $C_0 = \mathbf{a23.6.10.2969H}$

The 2969 words of C_0 , in the order of the source listing <https://aeb.win.tue.nl/codes/cwc/d6/a23.6.10.2969H>, thirteen to a line. Each is a six-digit hexadecimal mask on the 23 points, bit i (counting from the least significant, $i = 0$) standing for point $i + 1$. Positions referred to elsewhere in this note are 0-based indices into this list, read left to right and then down. The SHA-256 digest of the concatenation of these 2969 tokens, in this order and with no separator, is the digest given in §1.

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2822BD 283BB0 2843B6 284F89 287AC1 287C98 28863E 288AAB 288BC6 28935C 289CE2 28A14F 28A5D1
28B52A 28C6D2 28CC39 28E4E4 28EA70 2901AF 2906CD 29196A 291C33 29319C 293361 293626 293C4C
29459A 294CE8 295259 295534 29588B 29630D 298E58 299703 29A694 29A82E 29D065 29D2B0 29DB08
29E815 2A0697 2A074E 2A09B3 2A1C56 2A1E49 2A21E9 2A2A47 2A2EE0 2A5163 2A5A94 2A615A 2A6A8A
2A7506 2A8955 2A9E30 2AA534 2AAB48 2AB4C8 2ACE44 2AD590 2AE423 2B0176 2B083D 2B14D1 2B16A8
2B23A4 2B246A 2B28D4 2B4750 2B59C0 2B8E81 2B9149 2B9898 2BC60A 2BE8A0 2C04F3 2C2137 2C2C4B
2C3554 2C5938 2C609B 2C720E 2C7C22 2C859C 2C9287 2C964A 2CA362 2CA929 2CC0AE 2CD0D8 2CEE80
2D03D2 2D0FOA 2D15A2 2D1916 2D1A8C 2D2A32 2D30E8 2D6151 2D7601 2D9870 2DAA44 2DC168 2DC704
2DC892 2E0CA9 2E1F80 2E21C6 2E22A3 2E2E18 2E4515 2E46B0 2E6249 2E8C0E 2EA06C 2EC132 2ED228
2EE181 2F04E4 2F2C05 2F3430 2F4921 2F601C 2F7082 2F8942 2F9124 301AC7 3026D3 302B2B 3030DE
303374 30351B 304D55 304ECC 305467 305AF0 305D92 30628F 308787 309C3C 30AD25 30B546 30C3E2
30D0EC 30F603 30F988 3107B4 311F11 31274A 3132A3 3149C6 31538C 3160BA 316523 316B60 316E82
317630 318B2C 319076 31921B 31A097 31A271 31AE09 31B581 31D6C0 31D905 320277 3209AD 320CF2
32155C 321FOA 322BD0 324B16 3264CA 32A0E6 32A11E 32A469 32C709 32DA0C 330593 33061D 33085E
3311AA 3312D2 331A89 3322C5 333035 3340F4 336213 336849 33890B 3392A4 33B043 33D118 3400EF
340F38 3411BC 34160F 341A6C 3423C9 343A19 3445B1 34581E 346654 3468E8 3489A3 3490D5 34AC1A
34C24D 34E12A 34FA40 35089B 350B54 352C34 3534D0 353528 35384A 35446A 3550C9 3585C4 359192
35A226 35A318 35C03C 35C053 35CA88 3606AA 3642D8 365381 365434 367608 368839 3698E0 36A20B
36A88C 36C164 36D04A 370D22 370E90 371238 371C41 373106 376130 37A640 3805E6 380CD9 382A5C
382B85 3834AC 383C83 3854D4 386C61 387191 38918D 389453 38984E 38B0F0 38C1B8 38D28A 38E648
38F026 39015B 391D88 394037 39489C 395C06 39606C 3962D0 396584 398A07 398CA4 39960C 39A4C2
39B920 39E00B 3A0678 3A0A1B 3A0C27 3A1871 3A2598 3A3909 3A4724 3A5348 3A5612 3A5CA0 3A816A
3A82CC 3A85A1 3A8896 3A9464 3AAE02 3AB214 3AC02D 3ACB80 3AD803 3B0AA2 3B3540 3B4C11 3B7A00
3B81D0 3C08B5 3C09CA 3C2D0C 3C302B 3C3252 3C415C 3C423A 3C448D 3C8549 3CD121 3CE211 3D022D
3D0C52 3D1505 3D4845 3D50A4 3D6302 3D82E0 3E1322 3E1944 3E2A24 3E3890 3E410B 3E45C0 3E8A41
3E8D10 3F0388 3F0868 3F8085 3FD400 40197E 402EEC 4034D7 403B99 404677 4047BC 4049DD 405AE6
4061B7 40655B 406AF1 406B3A 40724F 407768 407C2E 408AD7 408BF8 408CBE 4095EC 4099CB 409D27
409E4E 40AE59 40B70B 40BB2C 40BCA9 40C7CA 40D1D6 40DC1D 40E0EE 40E52D 40E687 40F331 40F958
41053F 410DF4 410E9D 411B2B 411C6D 4127D8 412C7A 412F13 4138E3 4139CC 413E8A 414B6C 414CCE
4152DA 4154B3 415719 4163E2 41670E 41698B 416E29 41703D 417654 4186E3 4189AD 41925D 419372
41AB4A 41B0BA 41B195 41C07B 41C55C 41CA35 41D383 41D6A4 41D8C5 41D90E 41E29C 41E8D2 41F164
41F443 41F588 41FA22 4203F6 420B3D 421CEA 4226B3 4229DA 422D36 4230AF 423D43 4242EB 42456E
424CAD 424F07 4255D8 425696 425A55 426365 4268C7 426DE0 426E4A 427471 4278B2 427F10 428DC6
4291B3 42999C 429A36 429D70 42A2CD 42A467 42AAE2 42AF21 42B354 42B8D1 42C25E 42C395 42CAD3
42C8F4 42D06D 42D3E0 42D623 42D70C 42DD81 42E0B9 42E172 42E634 42EA13 42F107 42F28A 42F4C4
43094F 430AE5 430DA3 430E2E 4310F9 431653 4318D6 432387 432BA8 432CC9 432F44 43321E 4334B4

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433A31 4340BE 434333 4346F0 434D52 43528D 435346 435827 4361D4 436259 436417 436538 43706A
 4371A1 437AC0 43829B 438C39 439325 439487 439F02 43A07C 43AD90 43B1C2 43B660 43B909 43C50B
 43C645 43C9C8 43CA86 43D151 43D238 43E4A2 43E861 43F814 4409E7 440F8E 44131F 441BD4 441CB5
 4426CB 442D69 442E27 4431F8 443933 4444FA 444B93 44516B 445B25 445DA8 44661D 446974 44738C
 44817D 44863B 44968D 449BA2 449E61 44A2B5 44A39A 44A4DC 44A81F 44B46A 44B586 44B945 44C32E
 44C44F 44CCA3 44CF09 44D0F1 44DE90 44E1C9 44E263 44E6A8 44EE44 44F216 4505D3 45065E 450B71
 450EB2 4511AE 4516E8 45233C 4524B9 452996 452C55 453266 453349 45340F 4548E9 454E43 455057
 45549C 455570 4562C5 456CA4 4572B0 45781A 458B07 459934 459C13 45A12B 45A9E0 45AE81 45B618
 45C1B8 45D429 45DB40 45E911 46049F 460AD9 4611CD 461E83 461F48 462257 462678 4632E1 46332A
 463A0D 463C64 4646CC 46513C 465272 4662A6 46642B 466781 466D0C 467095 46834B 4684E9 468B64
 468D2A 469B11 46A1AC 46A60E 46C037 46DC42 46E2D0 46F920 470BC2 471398 47143A 471515 472925
 473950 474B14 474C91 4750E4 477203 4780D5 47811E 478E50 479063 47944C 479CA0 47A232 47B284
 47C2A1 47C524 47D092 47E046 47E308 4804EF 4811BD 4816DC 481D59 481E17 48232F 482BC3 4832B6
 48335A 48343B 483CF0 48469B 48562D 485C63 485F0A 48647C 486B15 4870E5 4874CA 487E84 488B33
 488C75 489369 4896AA 489B85 48A517 48A9D4 48AE92 48B44D 48B88E 48B962 48C1AB 48C947 48CE26
 48D274 48D398 48E34C 48EBA0 48ECC1 48F093 4903BA 490679 490B56 491BE0 491CA6 49215D 49249E
 4929B1 492E34 493907 493E41 4942A7 49434B 4945E1 495878 495994 4961AC 496D48 49816E 49860F
 498ACC 498D8A 49911B 49943C 49A265 49A389 49A4E8 49AC23 49BB10 49C2D1 49CCB0 49D648 49D921
 49E036 49F419 4A017B 4A07D1 4A1267 4A189B 4A1CC5 4A239C 4A2B70 4A316C 4A3D88 4A3E22 4A4BC4
 4A52B1 4A5C34 4A7056 4A866C 4A89E1 4A8E43 4A92D2 4A9619 4AA48B 4AAB06 4AAC44 4AB035 4ACD18
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1638A9 2139A9 23B984 259B21 2431D3 08CA9E 04C796 065547 0A584F 0EC487 076586 25E18C 227817
05C09F 0F029E 3E2416 268C93 2605CB 2155C3 11454F 394643 3D8423 20DB32 02C98F 24E4C3 006CEB
0C61E3 166561 22C5E8 02CD63 201F63 648745 0CA70D 02EF84 20778A 2C65A8 0493D9 2E1199 2A51AC
20F1C5 1D0333 28133B 061A3B 165859 1A5551 20D751 005B79 225D19 0C5671 0C529D 264A1D 0ACB25
2C4365 1389B8 30C87A 382972 1C2D23 3E2701 26B30C 349714 00973D 0C9A35 261731 271552 6A151A
42BC1A 23AA92 2DA28A 0D8A53 2C025F 2A80DB 28A99A 24B88B 038FE0 1897E0 2CB01D 2CDC41 207F44
04BF0C 047AD8 0877D0 203E5A 047713 087A1B 20DDA4 08D8AD 10ECAC 1CE1A4 24CD4A 239C4A 208CCF
20BD49 10BAC9 05B744 098774 058A7C 24A2F8 2588EA 619A68 69A238 392439 098D1D 25845D 27483A
036F22 0B6564 07686C 172A0E 1B2706 385A29 21562B 055A6A 095762 08E19D 0AE12E 0BBA0C 0C39AC
0F622A 12B8B4 204BDA 2495F0 369503

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